

FALCON's Response to Delhivery's RFQ for Bhiwandi Warehouse Automation for Bag and B2B Parcel Sortation

December 8, 2022



www.falconautotech.com

Kind Attention –

**Mr Sunny Raja
Mr Nikhil Ummat**

M/s Delhivery Ltd

Offer Ref: F22-00259; Date 08.12.2022

Subject – Techno-Commercial Offer for Warehouse Automation (Bhiwandi Location).

Dear Team,

We are pleased to submit our Techno-commercial Offer in response to your RFQ of bags and large parcel sorter for your Bhiwandi mega hub.

As you will note, we have studied your requirement in great depth and, along with the information collected during calls with you, we have put together a detailed techno-commercial proposal laid out in various sections and sequenced to enable you to understand our proposal and to re-enforce our commitment to be your partner in this important project.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Sortation Technologies and references.

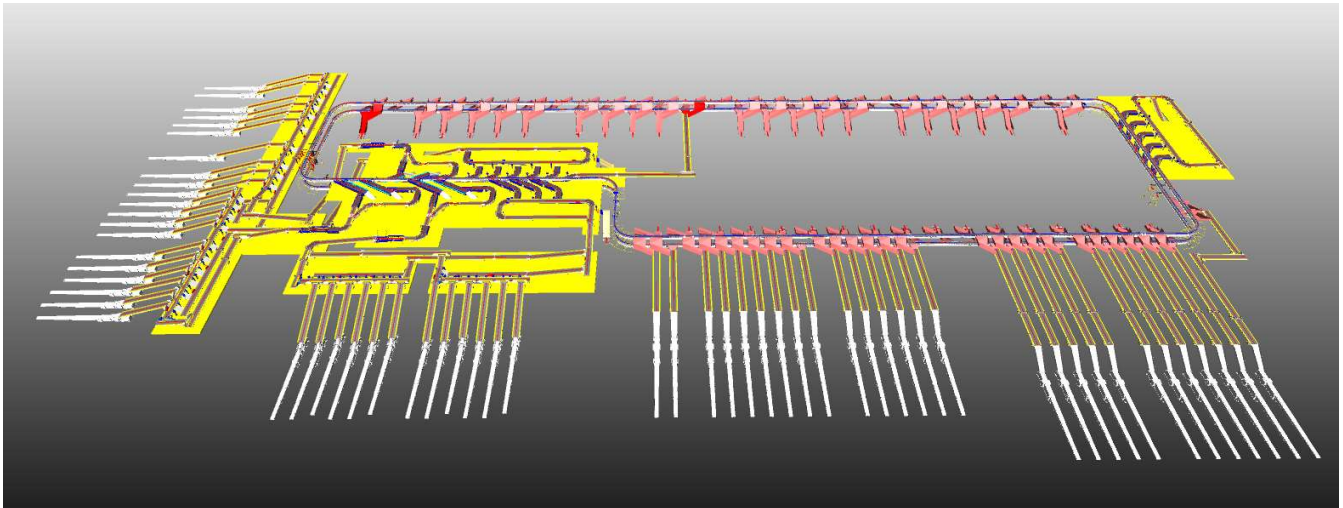
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through your award process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Adib Hasim

Manager – Sales & Solution

Double Deck Cross Belt Sorter ~ 17K TPH



Proposal Reference – F22-00259

**Falcon Autotech Private Limited
Plot No 308 & 309
Ecotech-1 Extension, Greater Noida
Uttar Pradesh- 201308
India.**

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Mob - +91 9711 619998
Adib.hasim@falconautoonline.com**

Contents

1. Glossary	6
3. Falcon's Experience and Achievements in Sortation Space Globally	17
4. Reference projects	18
4.1 Project 1- (Ecommerce Client).....	18
4.2 Project 2- (Client – Courier Express Parcel).....	19
4.3 Project 3- (Client – E-Commerce).....	20
4.4 Project 4- (CEP Client).....	21
5. Handled Parcel Spectrum for your Solution	22
5.1 Bags or Box size loadable on the sorter	22
5.2 Parcels to be loaded on Sorter shall have the following characteristics:	22
5.3 Parcel not loadable on the sorter	23
6. Proposed System Description.....	24
6.1 Objective	24
6.2 Summary of the System	24
6.3 Main benefits of the proposed solution	26
7. Layout View.....	27
8. Proposed System Capacity_ Calculations	28
8.1 Sorter System Capacity	28
9. System description	29
9.2 Feedlines:.....	33
Weighing Conveyor.	33
9.3 Main Loop.....	35
9.4 Output Chutes	36
10.2 Cross Belt Sorter -	38
10.3 CBS Carrier	38
10.3.1 Servo Roller.....	39
10.3.2 Chassis.....	39
10.3.3 Carrier Wheels	40
10.3.4 Non-contact based linear motor drive	40
10.3.5 Power Transmission.....	41
10.3.6 Data Transmission	41
10.3.7 Carriers positioning System	42

10.4	Scanning & Sensing on Main Sorter Loop	43
10.4.1	Automatic Barcode Scanning.....	43
10.4.2	Parcel Positioning System (Used for Physical & Virtual Centring)	43
10.4.3	Empty Carrier Detection System.....	44
10.5	Belt Conveyor.....	45
10.6	Plastic Modular Belt Conveyor.....	46
10.7	Powered Belt Merges & ZTR Curves (Pic for ref.).....	47
10.8	Roller type aligning Conveyors	47
11	Electrical System.....	48
11.2	Reference Picture of Power Distribution Panel.....	48
12	Controls Architecture (Reference Only)	50
13	IT Architecture (Ref.).....	50
14	High Availability Server Specifications	51
15	Falcon's WCS CONTROLIT.....	52
16	Proposed System Technical Details	53
	Technical specifications of Sorter.....	53
16.2	Mechanical Equipment- Bill of Materials	55
16.3	Electrical Equipment	57
16.4	Control System.....	57
17	Key Components Make.....	58
18	Automation Management System (AMS).....	59
a.	Issue Logging	59
b.	Preventive Maintenance Check List.....	60
c.	Predictive Maintenance	60
d.	Training Portal	61
e.	Operational Analysis	62
19	Commercials	62
20	Payment Terms	68
21	Warranty Period	69
22	Exclusions	70

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	DBO	Damaged Barcode
4	ICR	Intelligent Character Recognition
5	MEZZ	Mezzanine
6	LIM	Linear Induction Motors
7	ECDS	Empty Carrier Detection System

8	AC	Alternating Current
9	DC	Direct Current
10	PLC	Programmable Logic Controller
11	IT	Information Technology
12	BOQ	Bill Of Quantity
13	I/O	Input/ Output
14	PDP	Power Distribution Panel
15	PC	Personal Computer
16	UPS	Uninterrupted Power Supply
17	CBS	Cross Belt Sorter
18	DNF	Data Not Found
19	LGM	Logic Mismatch
20	RTVC	Real Time Video Coding
21	VDS	Volume Distribution System

2. Company Profile

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

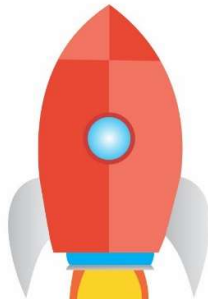
OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

COMPANY HISTORY



2021 ○

1. Company expands its manufacturing footprint by 4X.
2. Team size continues to grow, with 400+ on-roll employees.
3. Orders booked for some of the largest projects being executed in India.

○ **2018** 

1. Company sets up its first Sorter in Europe.
2. Company announces representation in North America.
3. Company ranked amongst the top 20 global players in the sorting automation market.

KEY STATISTICS

**04**Product Lines

**400+**On-Roll Employees

**215+**Engineering Team

**~150,000 Sq Ft**Manufacturing Space

**20+**Tier A Clients

**15+**Countries with Live Installations

**1700+**Total Installed Systems Globally

**35**Representations in 4 Continents and 35 Countries

**1 Million+**Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS

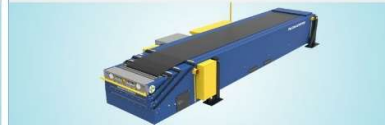


OUR PRODUCT PORTFOLIO

04

CONVEYOR AUTOMATION SOLUTIONS

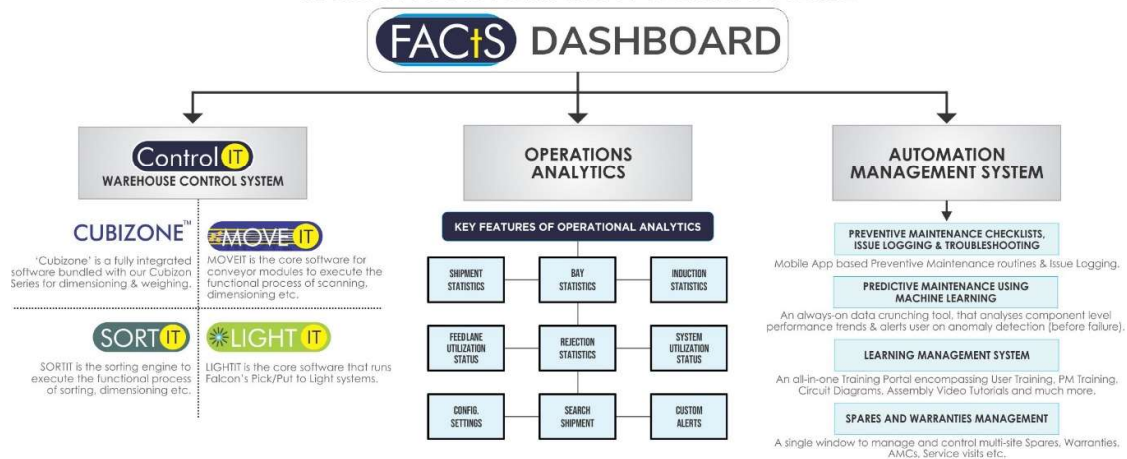
BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor



05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



— ADD-ON MODULES —



Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



Deep Learning Image Classification

DLIC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.

SOME OF OUR KEY PROJECTS

Falcon Autotech has pioneered a variety of unique Warehouse Automation solutions and has been successfully running several automation projects such as, high-speed Sortation Systems, static and online Weighing & Dimensioning Systems, Pick/Put to Light Systems, complex Conveying Systems etc. for E-Commerce, Courier, Express and Postal, FMCG, Cosmetics, Food, Pharma, Automobile, and Logistics Industries.

01



A high-speed Cross Belt Sortation project with one of India's leading E-commerce players for their high-volume site (>500k sq. ft. area). The solution is designed with a Volume Distribution System for handling the bulk volumes, Integrated Pre-Sorters, 28 automatic inducts with automatic weighing, barcode scanning, dimensioning and image capture capabilities. Falcon's Loop Cross Belt Sorter is designed to sort 27,600 parcels per hour and sort directly into 410 bagging chutes. All bags are closed sealed, scanned and taken away with the help of takeaway conveyors.

02



A complete end-to-end automation solution designed for a Pickup Processing Centre including Automated Shipment Unloading and Scanning, Automatic Conveying of shipments, Automatic Shipment Sortation at 24000 parcels per hour with automatic DWS, sorting into a total of 1080 end destinations with the help of primary and secondary sorting (using PTLs). This system is designed with a unique, proprietary feature provided by Falcon Autotech to handle the irregular parcels on the machine. This solution also consists of Bag Conveying, Automatic Bag Sortation at 3600 parcels per hour, sorting into 38 routes and Automated Loading of bags/shipments into the vehicles.

03



A high-speed Cross Belt Sortation solution designed for India's Largest E-commerce player for one of their high-volume sites. This solution caters to a throughput of 24000 parcels per hour, sorting directly into 100 end destinations. This solution is also designed with Falcon's Conveyor Automation modules for efficient material movement.

04



A complete end-to-end warehouse automation solution with one of India's largest Fashion Retail companies. The solution comprises of Automatic Unloading, Automatic Goods Receipt & Reconciliation, Put-away, Picking and Packing automation delivered through Conveyor automation modules and Vertical Reciprocating Conveyors. It also comprises of Falcon's high-speed Cross Belt Sortation Systems delivering a throughput of 16000 parcels per hour, for primary sortation combined with Put to Light System for secondary sortation.

INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



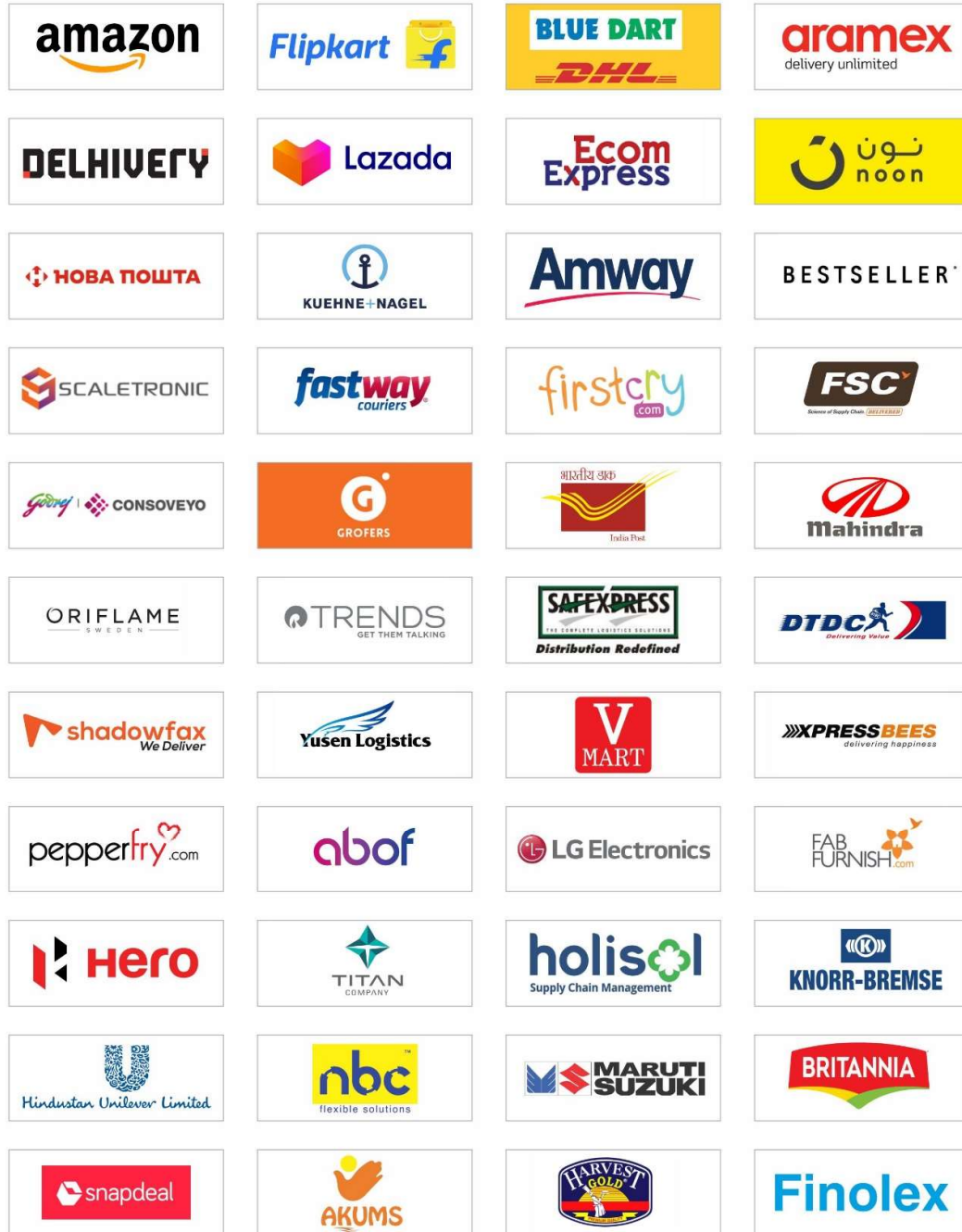
SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



MEDIA COVERAGE



Global Logistics Automation Market^{*} report published by Data Bridge Market Research, has featured Falcon Autotech amongst the **top 20 in global list of sortation systems manufacturers**. CNBC Awaaz, Awaaz Entrepreneur, featured Falcon Autotech as the **Fastest Growing Logistics Automation company** in India. Besides being featured in many important daily newspapers, we have also won the **Best Warehouse Automation Company of the Year** award at Express Logistics Supply Chain Conclave held in Mumbai in the Month of September 2015.

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Robotics powers festival sale impact on productivity for e-commerce brands

JESSE JOHN

October 26, 2016: Between Amazon's 'Great Indian Festival', Snapdeal's 'Utsav Divali' and Flipkart's 'Big Billion Day' sales, all eyes were on deals and wins earlier this month. However, what's common for most brands expecting large impact sales seasons is the need for operational efficiency. Falcon Autotech, an industrial automation solution provider in India, enables efficient fulfilment in the face of a surge of orders experienced by many online retailers during these high demand periods.

Vinay Bail, CEO, Falcon Autotech, explains, "It's the warehouses or fulfilment centres that see 2 to 5.5 times the regular volumes for large festive sale periods."

EFFICIENCY UNPLUGGED

THE FINANCIAL EXPRESS

How automation firm Falcon helping clients deliver products to workers

Industrial automation firm Falcon is helping its clients use robotic carts to pick, place and deliver products from specific locations to workers

Ever wondered how an e-commerce company fulfills thousands of orders everyday—picking up the product from the warehouse nearest the delivery address, packing it properly, labelling it with the right barcode and mandatory receipts, and then ensuring that it reaches the customer in minimal time? While an army of men and machines work round the clock to make it happen,

The Economic Times
SmallBiz
Home News SmallBiz Entrepreneurship SME Sector India & Overseas Marketing Strategy & Legal Money HR/Technology Startups
ET News SmallBiz Contact

Falcon Autotech looks overseas for growth

By Priyanka Jangam, ET Bureau | Updated Nov 01, 2016, 08:34 AM IST

PUNE: Falcon Autotech, a Noida-based maker of warehousing automation products, expects international operations to account for about a third of its revenue over the next few years, even as it sees demand for its

THE ASIAN AGE
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Hidden technologies changing our shopping experience

E-commerce sites finally solve the conundrum of sustainable growth and profitability

Are consumers ready for automation?
E-commerce sites have been looking for ways to improve their efficiency and reduce costs. One of the most significant ways is through automation. The use of robots and automated systems in warehouses and distribution centers has been on the rise. This is not only for picking and packing but also for sorting and labeling. The use of automation can significantly reduce the time and cost of these processes. However, there is a concern that automation might lead to job losses. But, many experts believe that automation will create new jobs in the maintenance and programming of the systems. The use of automation is also helping to improve the accuracy of the shipping process, which is a major concern for e-commerce companies.

DECCAN Chronicle

These tech oriented companies are making it smooth for industries to run

TECHNOLOGY IN OTHER NEWS

Industries still need to cross a lot of hurdles. Technology is playing role in every beat and is innovating to resolve the challenges. Here are few organizations that have made it better for industries in India. Not only have they saved time but are also saving a chunk of money for different industries.

Aaveg manages the entire activity chain including Employee transportation, Leasing to business rental and advising, managing and procuring the overall

PRODUCT CARGOTALK

Cutting edge tech for dimensioning & weighing

The highly accurate and versatile Cubicon-R THRU is equipped with roller bed base, has robust operations along with guaranteed zero-maintenance.

Cubicon-R THRU helps a company with warehouse storage space planning and shipment freight costs

Cubicon-R THRU is a highly accurate and versatile roller bed base, has robust operations along with guaranteed zero-maintenance.

^{*}Source: Report by Data Bridge Market Research

CLIENT TESTIMONIALS

“ Working with Falcon Autotech has been a very delightful experience for us, BLUE DART being a leader in the express logistics business, we expect the same kind of responsiveness, reliability, commitment & passion from our partners and Falcon has always delivered to their promise on all the solutions such as the Auto Sorter, Weight Dimension Labelling machine, Parcel Locker etc. All the technology interventions have been business critical for us.



Mr. Anil Khanna
Managing Director, DHL BLUE DART

“ Falcon is an organisation that has shown great agility and has helped us improve our service capabilities. Their sortation systems have helped us deal effectively with our growing volumes. We have found the Falcon team quite responsive to feedback and quick to incorporate changes into their systems that have helped build better, customized solutions for Delhivery.



Mr. Sahil Barua
CEO, DELHIVERY

“ Falcon Autotech has been efficient and proactive, particularly in meeting deadlines and their technical advancements. Falcon Autotech strike just the right balance between being a flexible, easy to deal with firm yet having all the key strengths and attributes. Every member of the team is honest, reliable, responsive, personable and very knowledgeable. I wish them good luck for all the future endeavours.



Late Mr. Sanjeev Saxena
Director, ECOM EXPRESS

“ I believe you guys earned it with your hard work and dedication. I believe the products you have designed demonstrate innovation and lateral thinking. Also the management has a flexible and the customer centric approach. It has been a delight to work with you and the team.



Mr. Amitava Saha
CEO, firstcry.com

3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolio in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 Million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

Born2Invest **Incas-SSI Schaefer merger creates leading player in Italy's intralogistics market**

Incas, the transportation management segment of the logistics automation industry is expected to have a higher CAGR. This is due to factors associated with the growing need to deliver goods in a timely manner and to reduce transportation costs.

Moreover, automotive will be the leading segment of the industry from 2018 to 2023, with an increasing demand for automated production systems and flexible logistics systems used in manufacturing and supply of vehicle fleets.

The MarkestandMarkets research report has identified SSI Schaefer as one of the key players in the logistics automation market worldwide during the forecast period, along with Dematic Corporation (US), Murata Machinery, Ltd. (Japan), Jungheinrich AG (Germany), and Falcon Autotech (India), among others.

This year's largest market for logistics automation is estimated to be North America, according to the research report. Regardless, Incas and SSI Schaefer's merger could help them compete in their main market.

(Featured image by DepositPhotos)

0 SHARES

Not secure | worldindustryinsights.com/30269/global-parcel-sortation-system-market-analysis-size-share-growth-trends-and-forecast-2019-2026-siemens-vanderlande-beumer-group



Global Parcel Sortation System Market Analysis, Size, Share, Growth, Trends, and Forecast 2019-2026: Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives

BY ANN.CASTRO ON AUGUST 12, 2019

The global "Parcel Sortation System market" research report is a pervasive research report that touches the most vital parts of the Parcel Sortation System market that is necessary to be grasped by a professional or even a layman. The market research report enlightens one regarding few of the important aspects such as an overview of the Parcel Sortation System product, the growth factors enhancing or hampering its development, application in the various fields, major dominating companies, genuine facts, economic situation, and geographical analysis. The research report endows the data concerning the dynamics that propel the growth as well as the supply and demand chain of the product on a global basis.



Click here to access the report: www.reportsbuzz.com/request-for-sample.html?repid=50822

The international and global stand of the Parcel Sortation System market is also briefly mentioned in the research report based on the performed statistical and thorough market analysis. The information mentioned in the research report gives a qualitative and quantitative view of the overall market. The statistical analysis of the market helps analyze the supply, demand, production, maintenance, and storage costs of the product. The data regarding some of the prevalent players Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives, Dematic, Interroll, Muratec, Invata Intralogistics, Viastore, Okura Yusoki, GBI Intralogistics, Equinox Mhe, Pitney Bowes, Falcon Autotech, OCM SRL, Solystic, Bowe Systec, Intralox are also detailed in the current case study.

4. Reference projects

Falcon has a strong legacy in **Warehousing Automation** solutions and takes pride in-

1. Expertise in delivering high throughput Sortation systems for bags and boxes.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

4.1 Project 1- (Ecommerce Client)

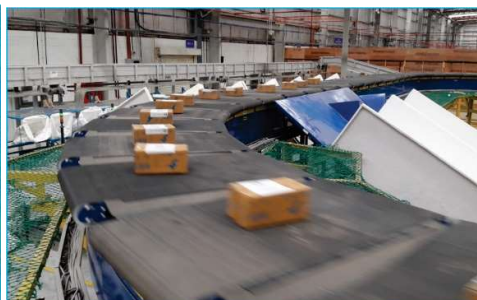
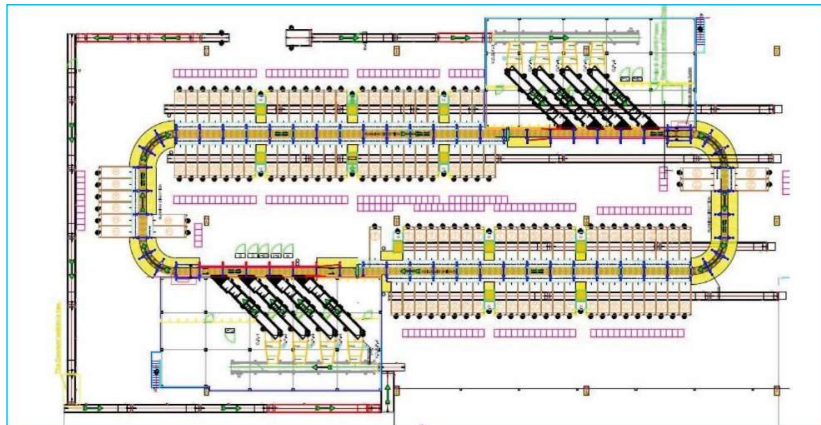
Use Case - Destination Sorting of Packed Parcels

Solution Specifications –

- Throughput: 16,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



4.2 Project 2- (Client – Courier Express Parcel)

Use Case - Courier Sorting of Parcels

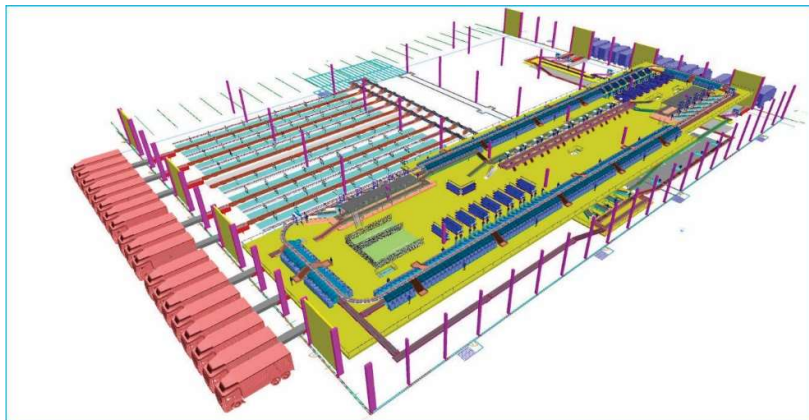
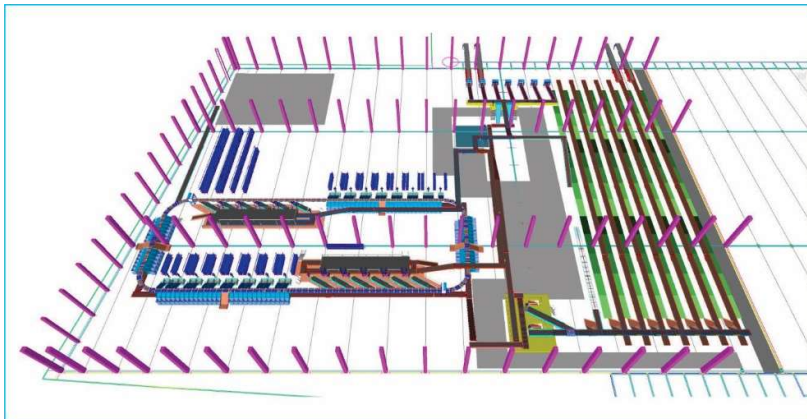
Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 394 Direct, 30 Secondary, 630 PTLs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Pre-Sorting
- Telescopic Belt Conveyors
- Bag Sorters
- PTL System

Layout -



Layout & Pictures

4.3 Project 3- (Client – E-Commerce)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

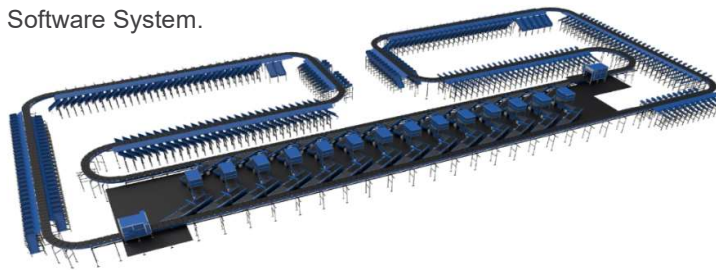
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



Layout & Pictures



4.4 Project 4- (CEP Client)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

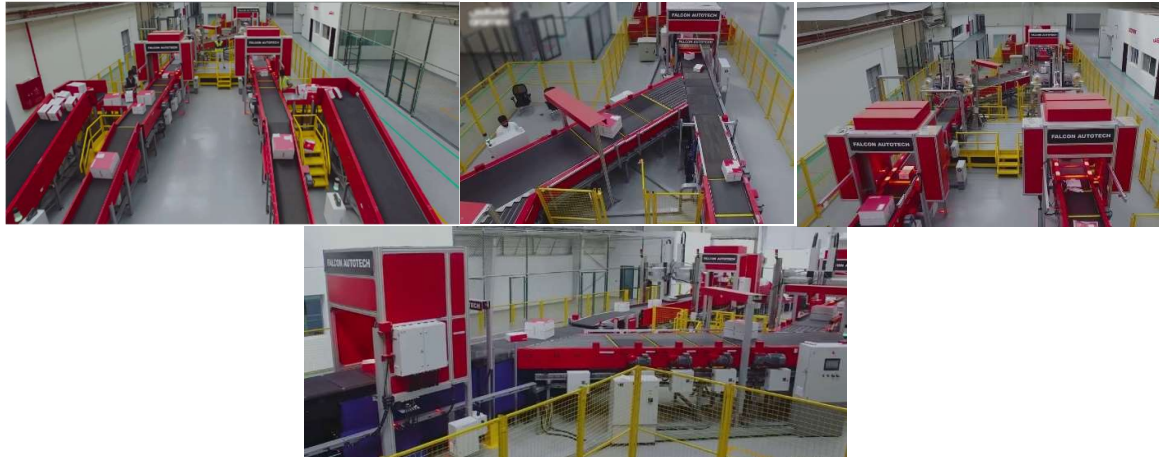
Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5. Handled Parcel Spectrum for your Solution

5.1 Bags or Box size loadable on the sorter

Max Length	mm	1000
Max Width	mm	800 (850 for Bags is allowed)
Max Height	mm	800
Max Weight	Kg	40
Min length	mm	150
Min Width	mm	150
Min Height	mm	50
Min Weight	Kg	0.2

Note:

- Weighing accuracy will be +/- 60 gms

5.2 Parcels to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content; otherwise, behaviour of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

5.3 Parcel not loadable on the sorter

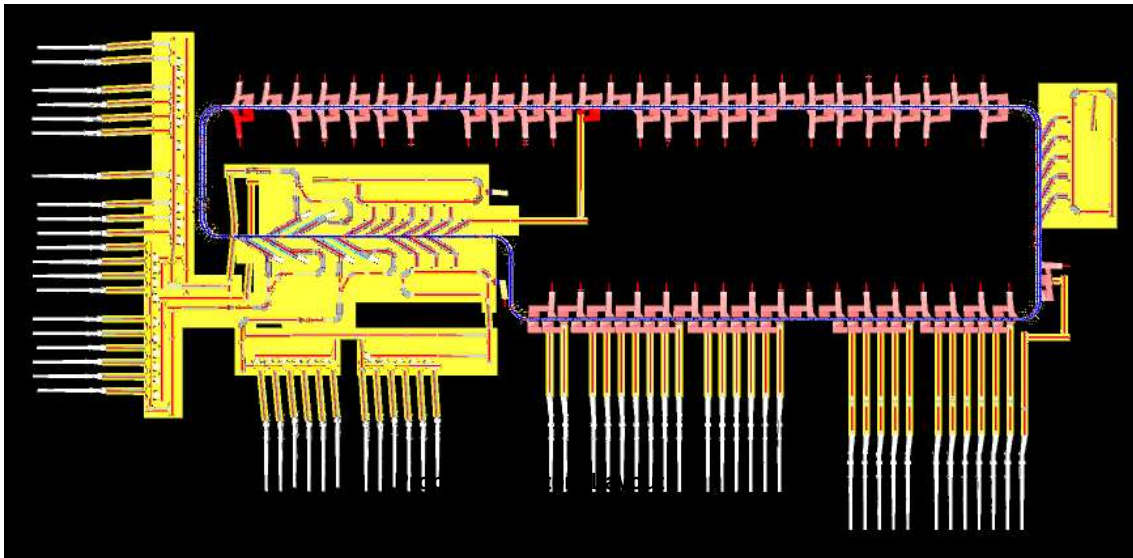
All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6. Proposed System Description

6.1 Objective

The purpose of this proposal is the design, manufacturing, installation, commissioning, testing and acceptance testing of a complete Sortation system to handle B2B Parcels and bags.



6.2 Summary of the System

Infeed System

- Infeed to the sorter is distributed into mainly 4 zones.
- 32 Docks in the South section constitutes Zone 1 & 2
- 12 Docks in the east constitutes Zone 3
- Output from the Parcel Sorter in the Northeast section constitutes the 4th Zone
- Zone 1:
 - Zone 1 constitutes of 10 Inbound Dock
 - Boxes and Bags are unloaded from the truck directly on the ten telescopic Belt Conveyors (TBC)
 - TBCs are connected to the common highway conveyors
 - 10 manual segregation stations with sliding chutes are positioned strategically on the highway line where operators will segregate bags and transfer them to the Bag Conveyor running parallel to the common highway conveyor
 - The main highway conveyor is equipped with mechanical aligners at specific intervals to ensure parcels and bags are aligned towards the manual segregation station for ease of access
 - Boxes on the Common Highway Conveyor will move to the singulators where boxes coming in bulk will be singulated automatically.

- Boxes will then move to the automatic induct zones of lower deck
- At the start of inducts, sensors will map the orientation of boxes based on which parcels will be either guided to the straight induct or diverted to the other induct via SWEDI.
- Bags from this zone will be conveyed through the bag conveyors to the lower deck bag VDS loop
- Operators at the bag VDS loop will manually induct the bag to the CBS via the 5 manual induct feedlines.
- Zone 2:
 - Zone 2 constitutes of 10 Inbound Docks.
 - Boxes and Bags are unloaded from the truck directly on the ten telescopic Belt Conveyors (TBC)
 - TBCs are connected to the common highway conveyors
 - 10 manual segregation stations with sliding chutes are positioned strategically on this highway line where operators will segregate bags and transfer them to the Bag Conveyor running parallel to the common highway conveyor
 - The main highway conveyor is equipped with mechanical aligners at specific intervals to ensure parcels and bags are aligned towards the manual segregation station for ease of access
 - Boxes on the Common Highway Conveyor will move to the singulators where boxes coming in bulk will be singulated automatically.
 - Boxes will then move to the automatic induct zone of upper deck
 - At the start of induct zone, sensors will map the orientation of boxes based on which parcels will be either guided to the straight induct or diverted to the other induct via SWEDI.
 - Bags from this zone will be conveyed through the bag conveyors to the lower deck bag VDS loop
 - Operators at the bag VDS loop will manually induct the bag to the CBS via 5 manual feedlines.
- Zone 3:
 - Zone 3 has 12 Inbound docks
 - Boxes and Bags are unloaded from the truck directly on 12 telescopic Belt Conveyors (TBC)
 - TBCs are connected to the common highway conveyors
 - 12 manual segregation stations with sliding chutes are positioned strategically on this highway line where operators will segregate bags and transfer them to the Bag Conveyor running parallel to the common highway conveyor
 - The main highway conveyor is equipped with mechanical aligners at specific intervals to ensure parcels and bags are aligned towards the manual segregation station for ease of access
 - Boxes on the Common Highway Conveyor will move to the singulators where boxes coming in bulk will be singulated automatically.
 - Boxes will then move to the automatic induct sets of upper deck
 - At the start of induct set, sensors will map the orientation of boxes based on which parcels will be either guided to the straight induct or diverted to the other induct via SWEDI.
 - Bags from this zone will be conveyed through the bag conveyors to the upper deck bag VDS loop
 - Operators at the bag VDS loop will manually induct the bag to the CBS via the 3 manual feedlines.

- Zone 4:
 - Zone 4 constitutes of the outputs of the shipment sorter
 - Positive Bags prepared in the Shipment Sorter will come through the bag take Away conveyors of the shipment sorter and will be connected to VDS System
 - Large Size Parcels which may come at the inbound of the shipment sorters will be directly fed in the VDS by a separate conveyor, the input of which is below the VDS Loop Mezzanine
 - Operators at the induct loop will manually induct the bags and boxes to the CBS via the 5 manual feedlines.

Sorting:

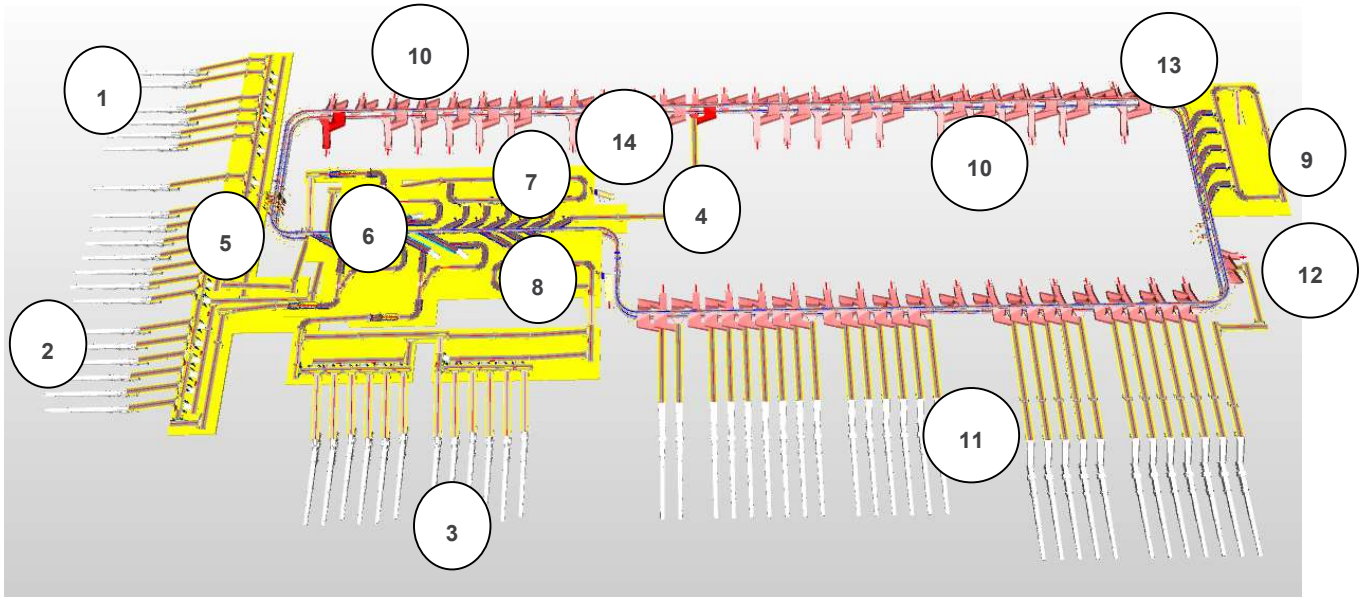
- There are 65 L-Type Collection Chutes and 27 Direct Takeout Chutes.
- Slow-moving location/Sector will be mapped to the L-type collection chutes and the bags and parcels sorted in this chute will be palletized and taken to the staging area for intermediate storage before line haul dispatch.
- The fast-moving locations will be mapped to the Direct takeout Chutes. These chutes will be connected with the outbound docks with conveyors, transfer chutes, and TBCs so that sorting and loading to the line haul trucks can be done simultaneously in real-time.
- All boxes inducted through the auto-induct feedlines will go through the 5-Side scanner and will be sorted as per the captured data.
- Bags will be sorted based on the data captured by manual scanning by hand scanner at the manual induct feedlines by the operator
- There are 2 rejection chutes.
There is one dedicated DBO rejection chutes which will deposit boxes on a slow-moving conveyor which will be re-inducted to the CBS through a dedicated manual induct feedline with manual scanning.

6.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput of ~17k boxes/ bags per hour.
2. Ability to handle entire range of conveyable Parcels with a sorting conveyor speed up to 2.3 m/s.
3. Low occupancy of floor space in the building using efficient Double Deck Loop CBS layout and chute design.
4. Narrow discharge centres for increased number of splits in limited space.
5. FALCON's CBS can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

7. Layout View



Legend

- 1 – Zone-1 Infeed Telescopic Conveyor – 10 nos.
- 2- Zone-2 Infeed Telescopic Conveyors – 10 nos
- 3- Zone-3 infeed Telescopic Conveyor – 12 nos
- 4- DBO re-induct take-away conveyor
- 5 – Zone 1 & 2 highway conveyor network for parcels and bags
- 6 – Zone 1, 2 & 3 auto inducts for parcels
- 7 – Zone 1 & 2 VDS loop with manual inducts for bags
- 8 – Zone 3 VDS loop with manual inducts for bags
- 9 – Zone 4 VDS loop with manual inducts for bags from B2B CBS O/p
- 10 – L-Type collection chutes, 65 nos
- 11- Direct Take-out chutes – 27 nos with conveyor and TBC to docks
- 12- Mixed bag sorting chute, o/p to be taken to CBS for shipment sorting
- 13 – Main Loop, double deck cross belt sorter
- 14 – Irregulars Take out Chute

8. Proposed System Capacity_ Calculations

8.1 Sorter System Capacity

The following table shows the Throughput Calculation for the Sorter

Upper Deck			Lower Deck	
Bag Inducts	3		Bag Inducts	7
TPH per Bag Induct	700		TPH per Bag Induct	700
Bags	2100	Bags	Bags	4900
Belts used	4200	Belts	Belts used	9800
Left Over Belts	9600	Belts	Left Over Belts	4000
Dual Belt Boxes*	15%		Dual Belt Boxes*	15%
Boxes Induct	7200		Boxes Induct	3478
Box Inducts	4		Box Inducts	2
TPH per Box Induct	1800		TPH per Box Induct	1739
TPH per Singulator	3600		TPH per Singulator	3478
Box TPH	7200		Box TPH	3478
Bag TPH	2100		Bag TPH	4900
Upper Deck TPH	9300		Lower Deck TPH	8378

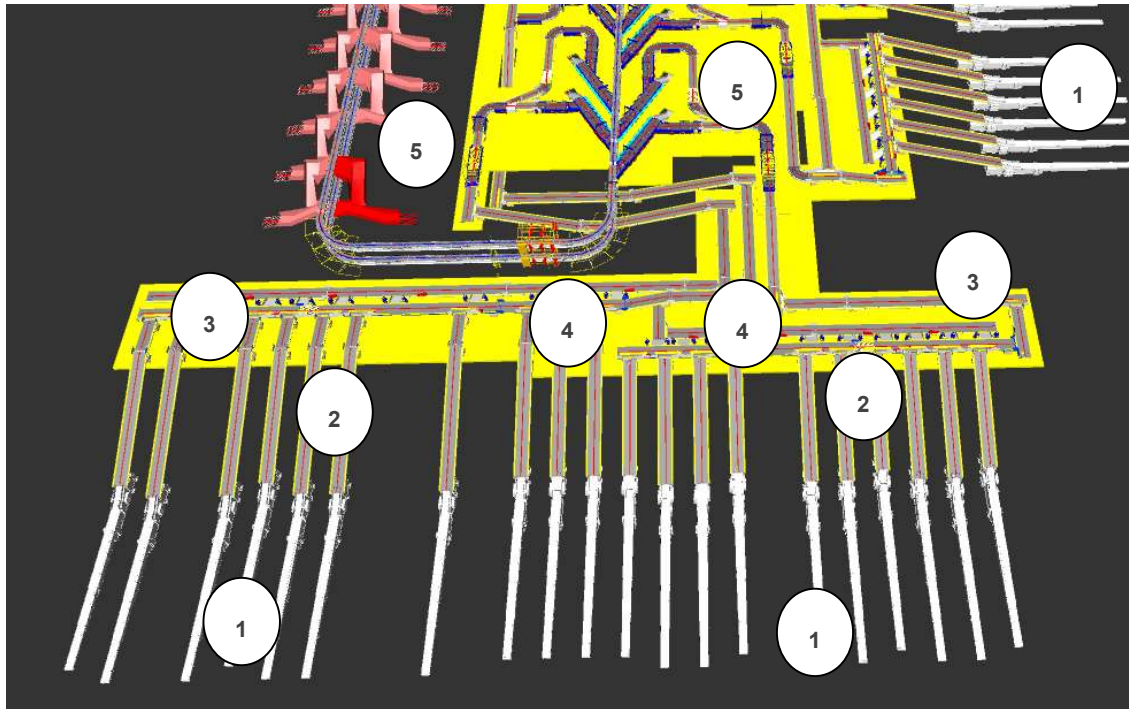
*Basis the analysis done on ~3 million Delhivery's B2B shipment dimensions.

Final TPH (Box and Bags)	
Total Boxes	10678
Total Bags	7000
Total TPH	17678

Final TPH (Lower and Upper Deck)	
Lower Deck	8378
Upper Deck	9300
Total TPH	17678

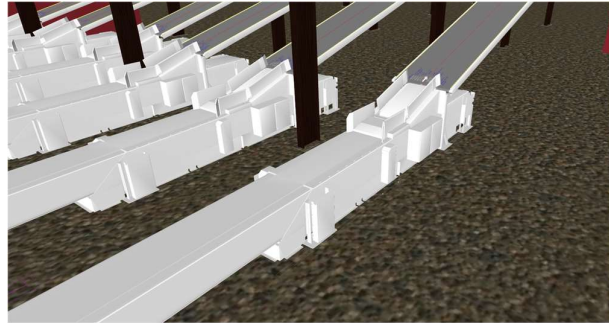
9. System description

9.1 Infeed System: Infeed System consists of below component:



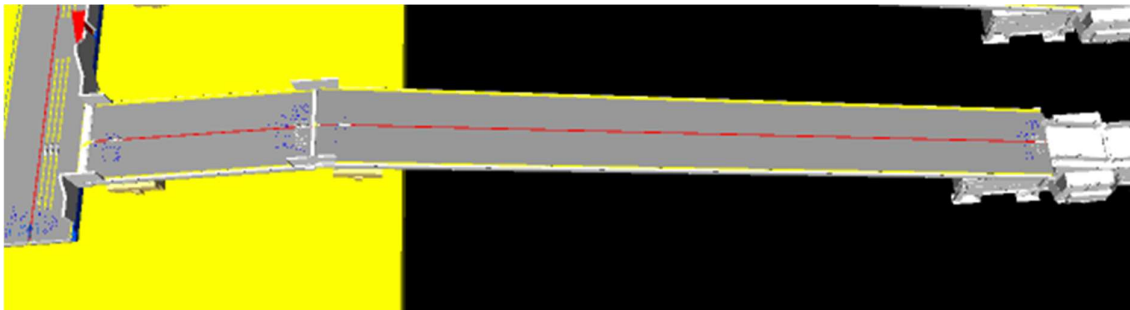
1. Telescopic Belt Conveyor with Early Pickup Unit
2. Inclined Conveyor
3. Highway Conveyor with mechanical and powered aligners
4. Manual Bag segregation stations with sliding chutes
5. Automatic Singulators for parcels

9.1.1 Telescopic belt Conveyor (with Pickup Module)



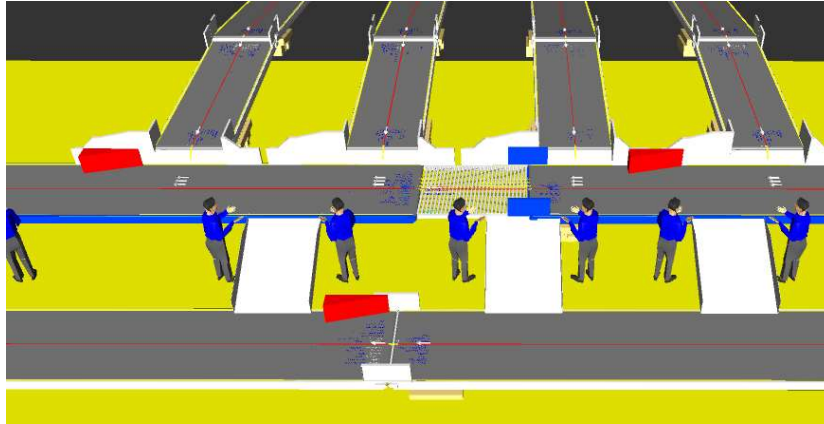
Specification	UOM	Remark
Base Length	Mm	~7000
Extended Length (Total Length)	Mm	~21000
Conveyor Belt Running Speed	m/min	Up to 20
Belt Width	mm	800
Hydraulic Feature	NA	Yes, up to 4 degrees <i>(Hydraulic Feature might not be available in TBCs with Early Pickup Unit)</i>
Communication	Mm	2 x RS243, 1 x USB2.0, Ethernet

9.1.2 Inclined Conveyors



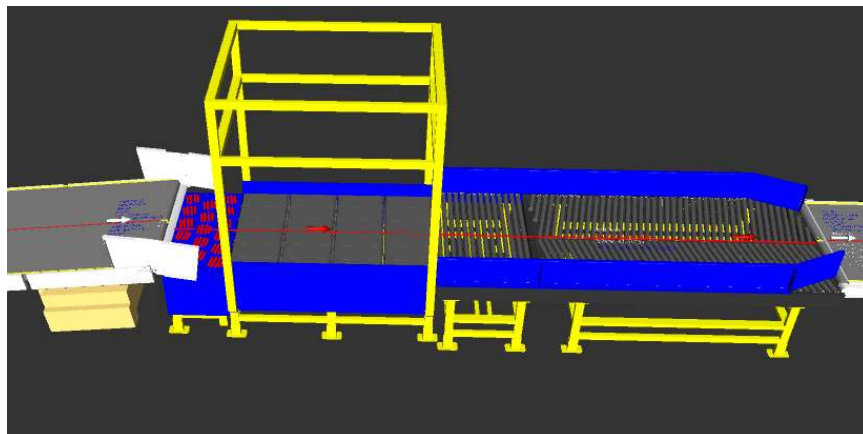
- Inclined belt conveyor from TBC (width 1400 mm)

9.1.3 Highway Conveyors with aligners



- Highway Modular Belt conveyor for parcels and bags. (Width 1400 mm)
- Powered Aligners at specific intervals to align parcels/bags to the operator side for bag segregation
- Mechanical aligners to avoid parcel pile-up and create space for Bulk dump from Incline conveyors

9.1.4 Singulators



- Automatic 3D Vision based parcel singulators
- A combination of multiple modules of roller aligners, zonal & split belt and swivel wheel mechanisms to segregate heap or jumbled up parcels to a steady flow of

singulated parcels. Client needs to ensure a 2D feed to the singulator by pacing human operators at the infeed for the singulator.

- Throughput – 3600 parcels/hr on average size parcel

9.1.5 SWEDI Diverter

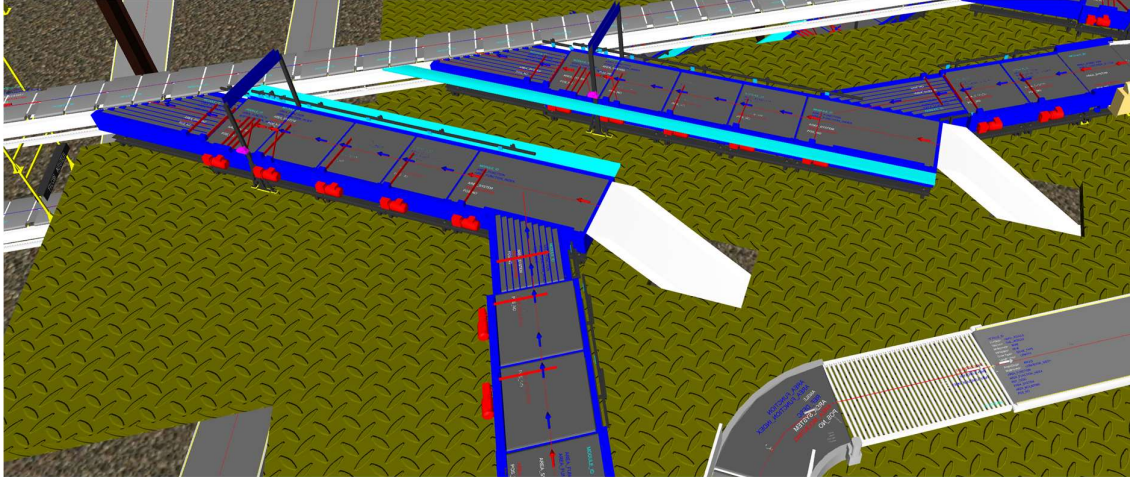
Swivel wheel module to divert and distribute singulated parcels between two Auto Induct Lines



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Type - Operation	Type	Sorting
Mechanism for Swivel	Type	Electric servo motor
Roller drive	kW	Powered smart wheel
Roller pitch (L x W)	Type	110 x 120
Max. product weight	kg	40
Throughput	PPH	3600
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

9.2 Feedlines:

9.2.1 Auto Induct Feedlines for parcels



- 6 Sets of Auto Inducts for Parcels equipped with Spacers, Buffers, Merge, MLGs and associated Controls.
- Each Feedline has 1 Weighing Conveyor
- Over-Sized parcel rejection chutes: In case overweight parcels are falsely inducted into system the receiving will run in reverse direction and drop these parcels in the rejection chute to avoid wrong induction in the CBS. These chutes end in a conveyor for manual handling downstream.

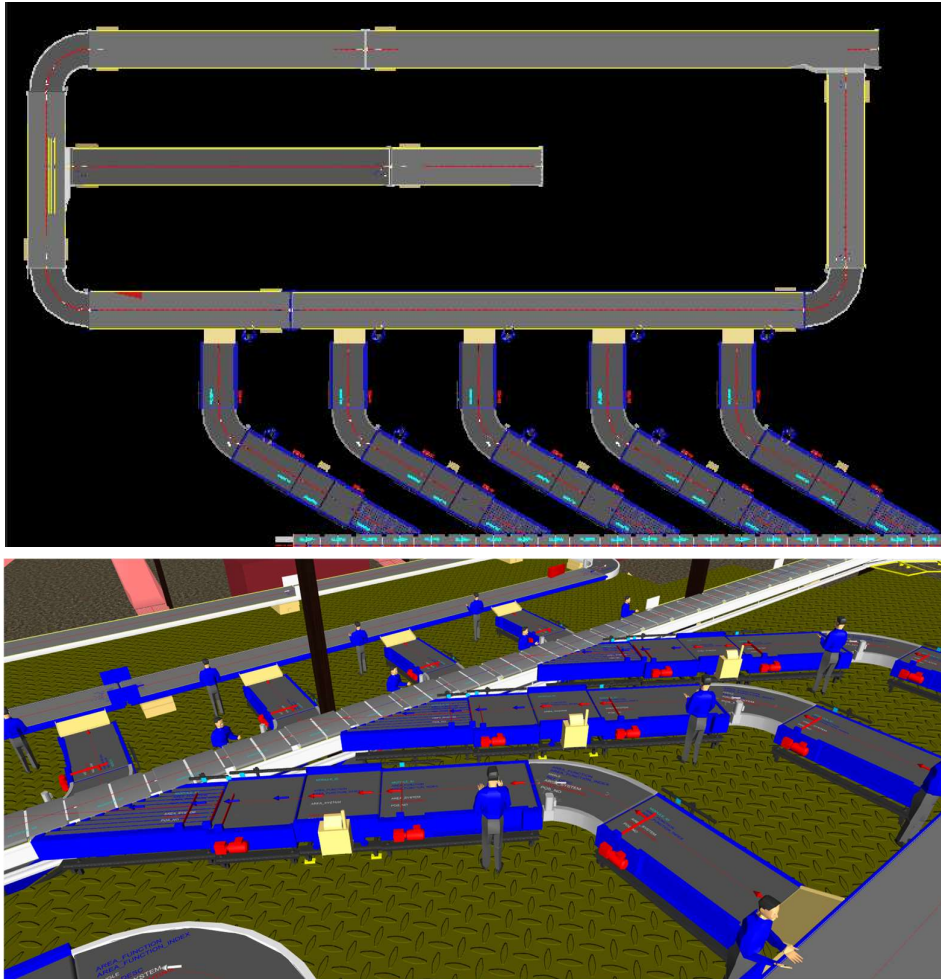
Weighing Conveyor.

Conveyors equipped with high precision Load Cells to capture the weight of Parcels.

(Bizerba/ Mettler Toledo)

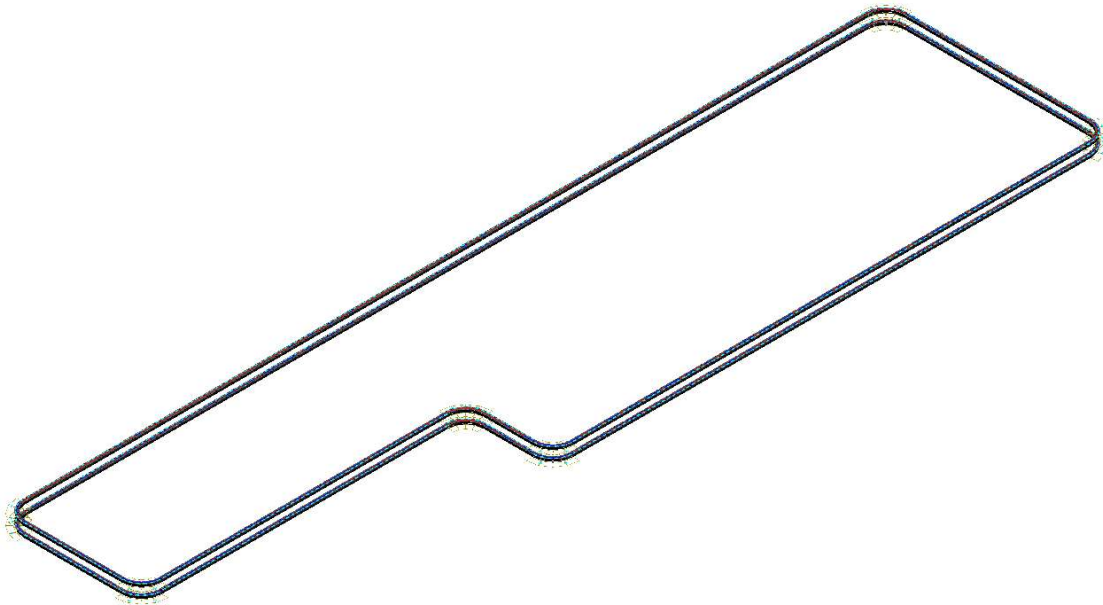


9.2.2 Manual Induct feedlines with VDS Loop



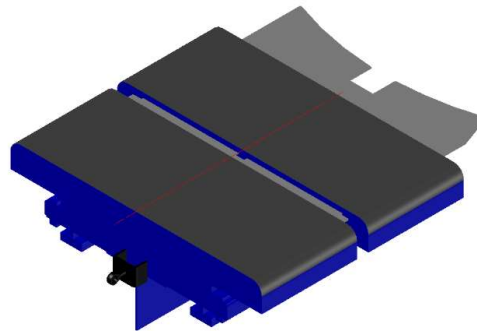
- Loop belt VDS conveyor for manual induction of bags and Exception parcels (width 1400 mm)
- Total 14 Nos. Semi-Automatic Feedlines (13 Nos. with transfer plates for dragging the bags from VDS Loop and 1 no. for DBOs)
- Each Feedline is equipped with Weighing Conveyors, Hand Scanners, Operator HMI and associated controls.

9.3 Main Loop



Main Loop

1. The proposed Double Deck Sorter will be installed on the Ground Level with the Feedlines on the Mezzanine. The First Deck of Sorter will run at ~3800 mm at 2.3 m/s speed and second deck at ~6300 mm at 2.3 m/s speed
2. We have provided Dual Belt Carriers with 1175 mm pitch.



3. As Bags or Boxes pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the bags or boxes reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the bags or boxes in the assigned chute.

9.4 Output Chutes

Two Types of Output Chutes are used in this solution:

- L Type Collection Chutes
- Direct Takeout Chutes

9.4.1 L Type Collection Chutes

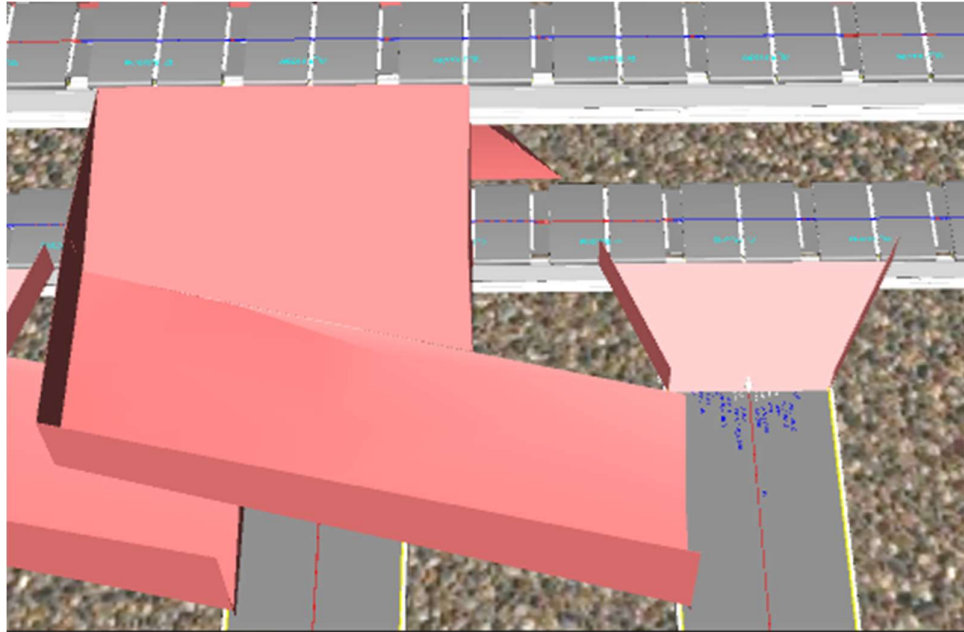


Collection Type Chutes

- These are collection type chutes where the Bags/boxes keep collecting into IDLER roller conveyor as the sorting continues.
- The user collects the boxes/bag from the roller conveyors.
- Duplicated destination from both the decks of the CBS

Chute IOs	Description
Single Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation.
Tower Lamp	To indicate the operator action on a chute
Push Button	To Suspend/Resume Sorting into a particular Chute

9.4.2 Direct Takeout Chute



- These are direct takeout chutes where the bags/boxes will be drop directly on the takeout conveyors.
- Takeout conveyors will drop the bags/boxes on the telescopic conveyor in turn which are connected to the docks for loading into trucks

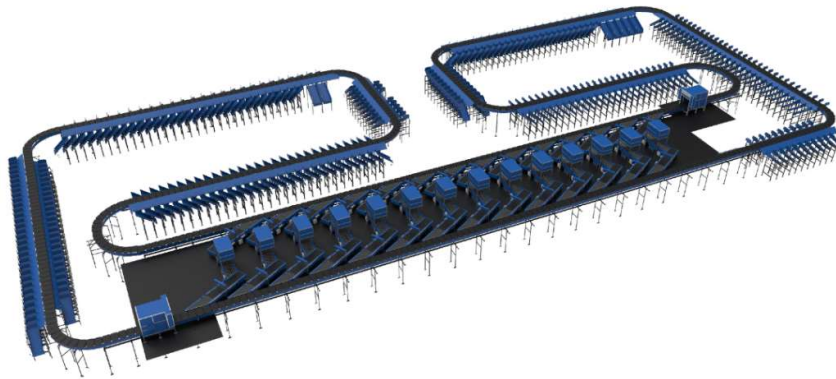
Chute IOs	Description
Single Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation.
Tower Lamp	To indicate the operator action on a chute

10.2 Cross Belt Sorter -

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels.

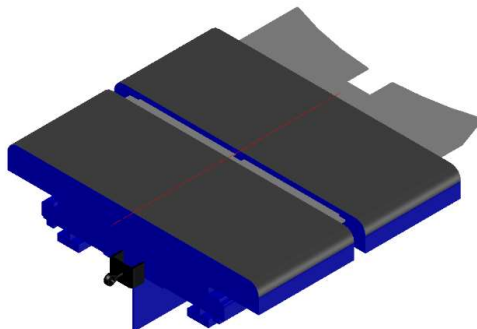
Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



Cross Belt Sorter

10.3 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners" and non-sortable parcel recirculation.



CBS Carrier

10.3.1 Servo Roller

- High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



Servo Roller

10.3.2 Chassis

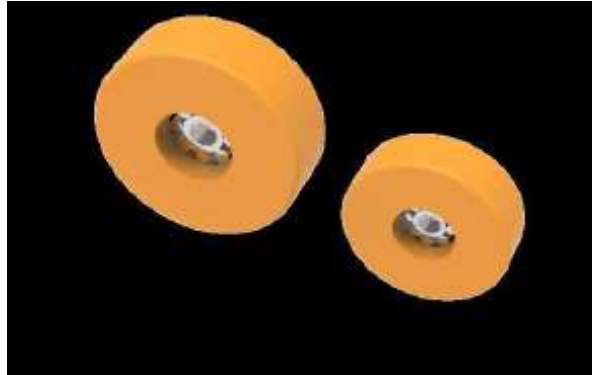
Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminum which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



Chassis

10.3.3 Carrier Wheels

- Carrier wheels that are thoroughly tested and proved for long life cycle



Carrier Wheels

10.3.4 Non-contact based linear motor drive

Extremely **High Energy Efficient** Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed



Linear Motor

10.3.5 Power Transmission

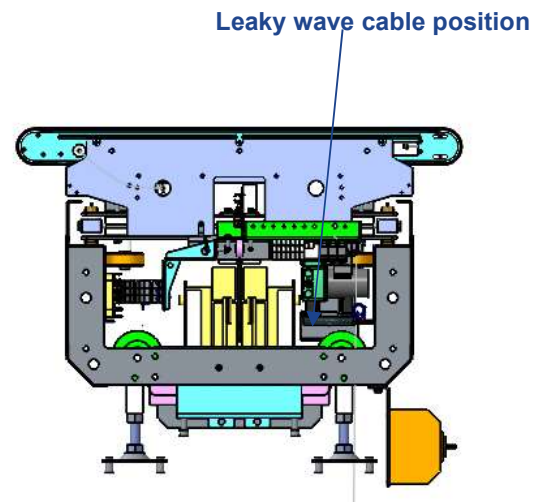
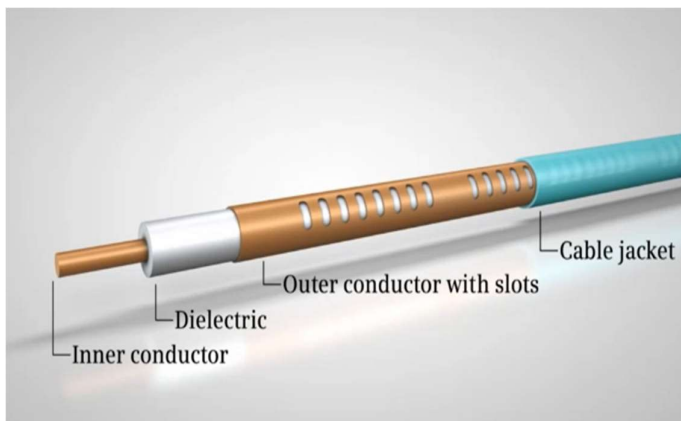
- Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



Power Transmission

10.3.6 Data Transmission

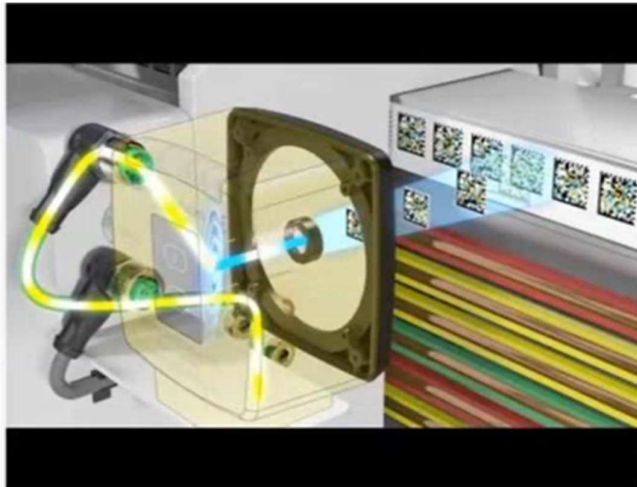
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



Data Transmission

10.3.7 Carriers positioning System

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



Carrier Positioning System

10.4 Scanning & Sensing on Main Sorter Loop

10.4.1 Automatic Barcode Scanning and Volume Scanning System

3 Side Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. This will enable 5 sided Barcode Scanning. Additionally, the system is equipped with Volume Scanning System that can capture the Parcel dimensions with ± 10 mm accuracy.

Handheld Barcode Scanners are provided to run the machine on Semi- Automatic Mode, if required



10.4.2 Parcel Positioning System (Used for Physical & Virtual Centring)

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



10.4.3 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the parcels on the empty carriers. (Camera based)



10.5 Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



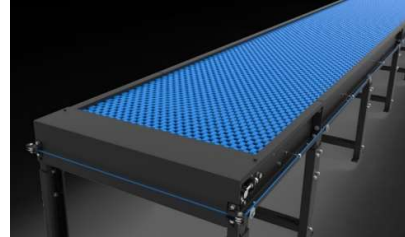
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Finish	Type	Matt Finish
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Motor Shaft	MOC	EN9 Shaft material,
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Type of Guides	Type	Steel guides at 50mm to 400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Drive	make	Lenze/Siemens/Omron/Allen Bradley
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

10.6 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make/Model No.	Make	Movex/XP254 CR
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Wear Strip Material	MOC	UHMW PE1000
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Drive Shaft	MOC	EN 24
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

10.7 Powered Belt Merges & ZTR Curves (Pic for ref.)



10.8 Roller type aligning Conveyors



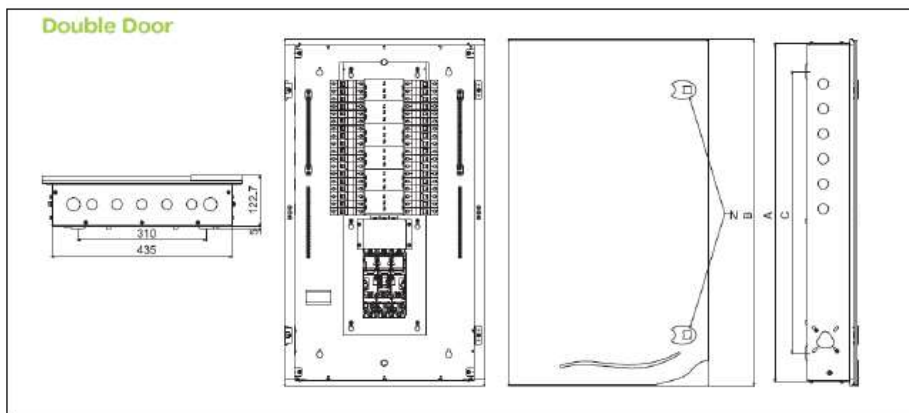
11 Electrical System

Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.

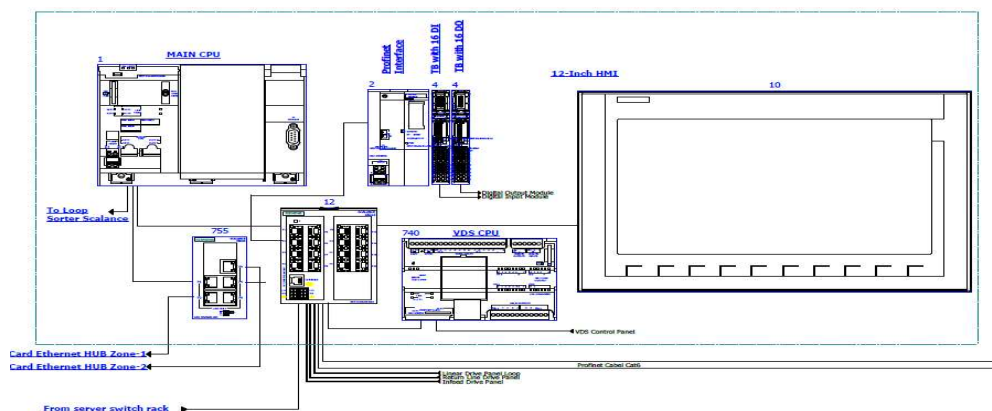
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

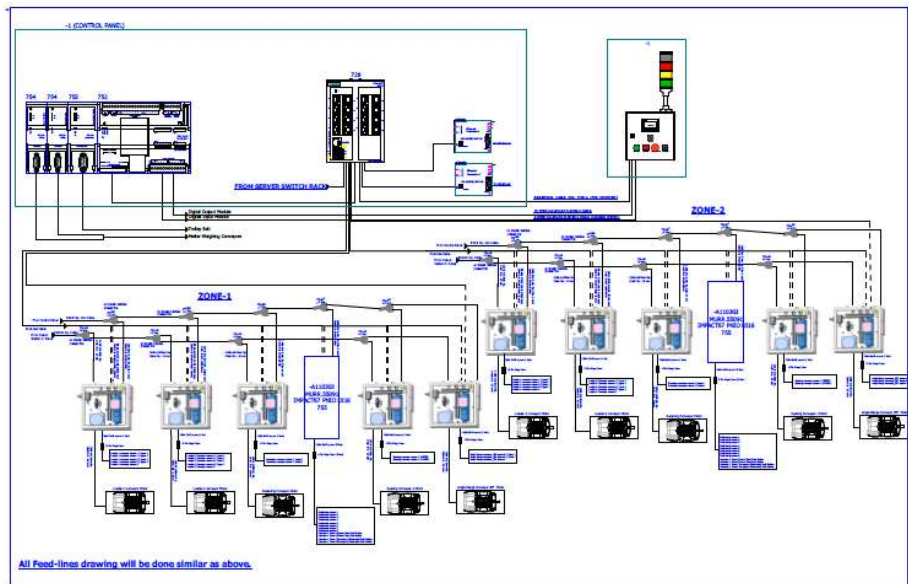
11.2 Reference Picture of Power Distribution Panel



Main Control Panel (Reference)



Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter. The engines will be coupled with a converter to improve consumption and reduce the carbon footprint. All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

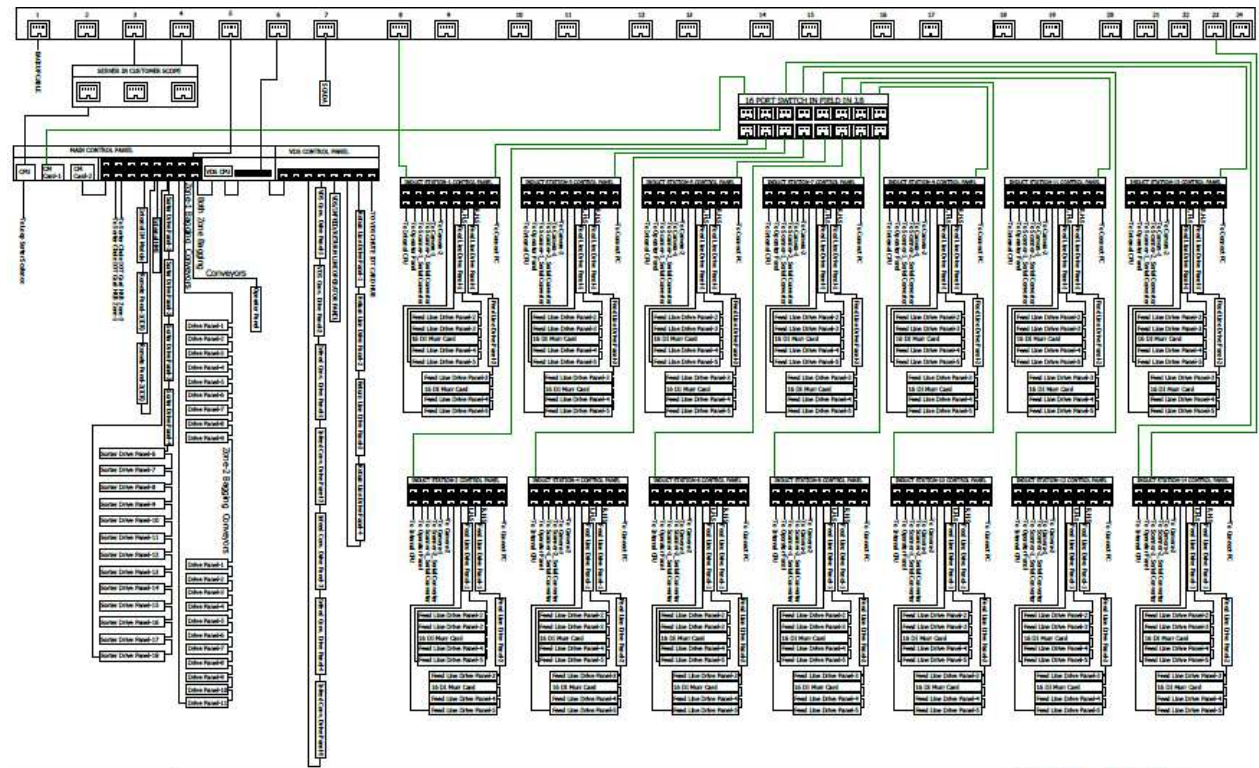
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform with optional Omron Platform for Conveyor Control. The entire system will be logically divided into zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet + Ethernet + EtherCat

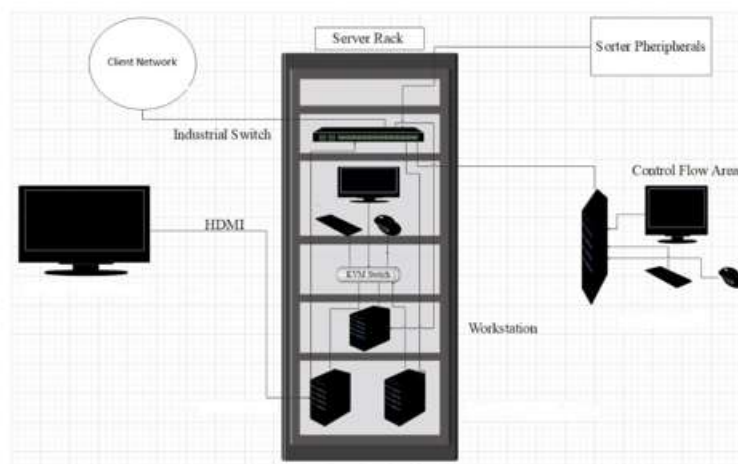
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

12 Controls Architecture (Reference Only)



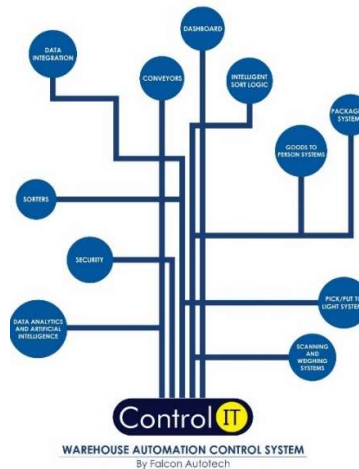
13 IT Architecture (Ref.)



14 High Availability Server Specifications

Product/Model	ITEM Description	Qty.
T440 (4210R) Tower 2P Server	Intel Xeon Silver 4210R 2.4G, 10C/20T, 9.6GT/s, 13.75M Cache, Turbo, 4x16GB DDR4 (16 Slots), 2x1.2TB 10K RPM SAS 12Gbps 512n 3.5in Hot-Plug Hard Drive upto 8 Drives SAS/SATA, 2x1Gbe LOM, RAID 0,1,5 (H330), 2xRPS Standard Inbuilt 495W,iDRAC9,Enterprise, 3 Years Pro-Support & Mission Critical/Security Bezel	2
T40-Tower 1P SERVER	Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 2x8GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1x1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3 Year Onsite NBD/ Extra LAN Port, 480 GB SSD	1
Synology_storage_DS720+	2 Bay Intel Celeron J4125 4-core 2.0 GHz, burst up, 2 GB DDR4 non-ECC Expand up to 7 drives with a DX517 expansion unit to 2.7 GHz,2 x built-in M.2 2280 NVMe SSD slots,3 Years	2
SSD for Synology	Seagate 1.92TB LE SSD	4
UPS	Microtek 1 KVA offline UPS	3
Dell Monitor	Dell 19.5" Monitor with HDMI And VGA Port	1
KVM Switch	Aten KVM Switch (4 PORT)	1
License	Windows Server 2019 Standard R2 OEM	3
Keyboard	Dell Keyboard Wired	1
Mouse	Dell Mouse Wired	1
Cisco Switch	Cisco Business 350-24T-4G Managed Switch (CBS350-24T-4G-IN)	1
Converter	DP to VGA Converter Cadyce Brand	1
Server Rack	AC Server Rack	1
5 Meter LAN Cable	5 Meter D-link CAT6 Cable	5

15 Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

16 Proposed System Technical Details

Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Dual Belt Regular
Dual Belt Feature	YES.
Sorter Speed in m/s	Up to 2.3 m/s
Sorter Loop Length (Lower Deck)	~512.4 m
Sorter Loop Length (Upper Deck)	~512.4 m
Length of the Sorter	Layout Attached
Width of the Sorter including Chutes	Layout Attached
Sorter Actuation Technology	Electric
Dual Belt Feature	YES
Chutes + Rejection	Layout Attached
Total outputs Location	Layout Attached
Height of Lower Deck Sorter	~3800 mm
Height of Upper Deck Sorter	~6300 mm
Track Material & Type of Design to move the Carrier trolleys	~Steel Track
Carrier Motor Make	Falcon
Carrier Pitch	1175
No. of Carriers (Both Deck)	~872 Nos.
Motor/Drive type	Linear Motor
Power Consumption	Will be specified during DAP
Single Operator Feedlines	21
No. of Barcode Scanners (ICR type for 5 sided reading).	2

Power Transmission	Over Continuous Bus Bar
Parcel Positioning System	2
Parcel Centering System	2
Empty Carrier Detection System (Camera Based)	2
Sorter Throughput (Both Deck)	~17000

16.2 Mechanical Equipment- Bill of Materials

Pos.	Qty.	Description	Value
FM1	1	Conveyor Package - Powered Belt Conveyors - Modular Belt Conveyors - Powered Turn Conveyors (30/60/90 Degrees) - Singulators - Telescopic Belt Conveyor (7+14 metres) - Idler Roller Conveyors (4 m) - Aligning Conveyor (4 m)	~1872 Metres (178 Conveyors) ~ 216 Metres (15 conveyors) 36 Nos. 3 Nos. 59 Nos 70 Nos 5 Nos
FM2	14	Manual Induct Feedlines Consists of - Receiving Conveyor - Loading Conveyor. - Weighing System - Dynamic Spacing System - Angle merge - Hand Scanners for Manual Scanning	1Set 1 Set 1 Set 1 Set 1 Set 1 No.
FM3	6	Auto Induct Feedlines Consists of - Weighing Conveyor - Spacing System - Receiving Conveyor - Buffer Conveyor - Angle Merge	1 No. 2 Set. 1 Nos 4 Nos 2 No
FM4	1	Falcon Lower Deck Sorter 1 Loop Cross Belt Sorter - Sorter Height - Sorter Length - Sorter Speed - Sorter Drive - Carrier Pitch Including: - Standard Sorter Supports - Empty Carrier Detection System - Product Centring System - Camera based Barcode reader - Dimension Scanning System - E-Stops	3800 mm ~512 m ~2.3 m/s Linear Motor 1175 mm

		- Hooters - Fencing - Netting - Overshoot Assembly	
FM5	1	Falcon Upper Deck Sorter 1 Loop Cross Belt Sorter - Sorter Height - Sorter Length - Sorter Speed - Sorter Drive - Carrier Pitch Including: - Standard Sorter Supports - Empty Carrier Detection System - Product Centring System - Camera based Barcode reader - Dimension Scanning System - E-Stops - Hooters - Fencing - Netting - Overshoot Assembly	6300 mm ~512 m ~2.3 m/s Linear Motor 1175 mm
FM6	1	Chutes - Transfer Chute - Sliding Chute - Irregular Collection Chute (Lower Deck) - Irregular Transfer Chute (Lower Deck) - Irregular Collection Chute (Upper Deck) - L Type Collection Chute - Direct Takeout Chute - Rejection Chutes - Chute Full Sensors - Tower Lights	27 Nos 32 Nos 2 Nos 3 No 2 Nos 65 Nos 28 Nos 1 No As per design As per design
FM7	1	Steel Works Operator and Maintenance Platform <u>Please note- The current Proposal does not include the Price of Steel Works and the same shall be discussed separately with yourself.</u>	~5763 SQM

16.3 Electrical Equipment

Electricals			
FC1	1	Consists of • Main Power Distribution panels • Main Control Panels • Feedline Control Panels • Conveyor Drive Panels • Sorter Drive Panels • Network Switches • Field Cabling • Cable trays • Remote I/O Panels	Included

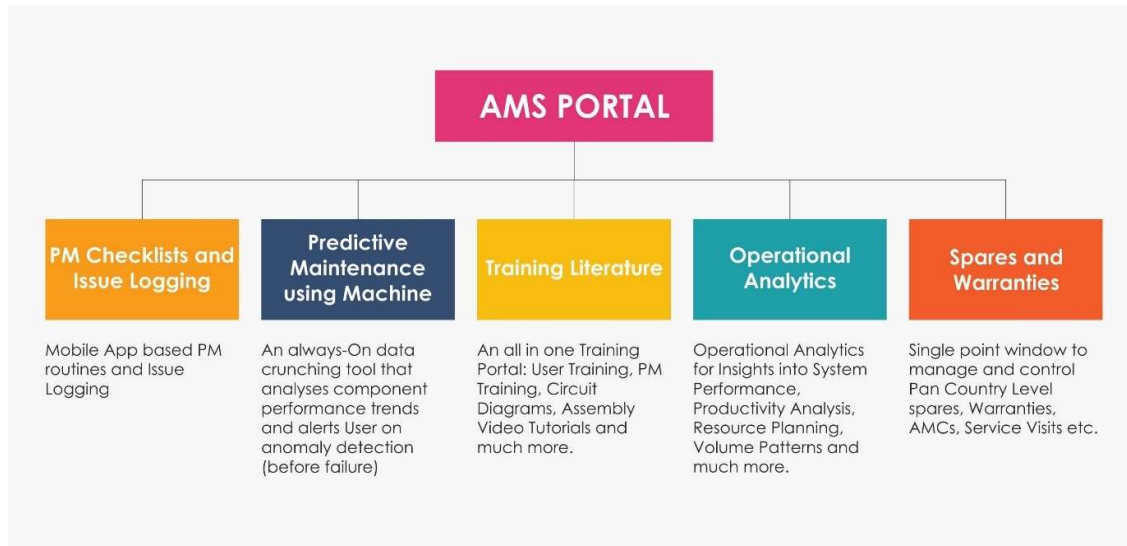
16.4 Control System

Components			
FC2	1	Controls and IT Infrastructure Controls • Weigh Scale Control • Barcode Scanner Controls • Dimension Scanner Controls • Conveyor Package Controls • Sorter Control • Induct Control • Smart Chutes Control IT • Control IT Software Package • High Availability Server Setup (Microsoft Clustering based) • Networking Switches	Included

17 Key Components Make

Items	Make
PVC Belts	Forbo/Derco
Rollers	Falcon
TBCs	Wangman/ Guanchao/ Pierski
Parcel Singulators	Wangman/ Similar
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Servo Motors: Lenze, Induction Motors – SEW/ Lenze/ Nord
Barcode Readers	SICK
Wireless Barcode Scanner	Honeywell/Motorola
Weighing Conveyors	Bizerba / Mettler Toledo
Volume Scanners	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens + Omron
Control Panels	Rittal/ BCH
MDR Rollers	Falcon/ Pulse/ Interrol
Power Transmission Systems	Vahle

18 Automation Management System (AMS)



a. Issue Logging

Issue Logging(T-214)

ADD ISSUE **OPEN ISSUES**

Problem Observed +

Reason +

Sub Reason +

Sub Sub Reason +

Corrective Action +

Time Taken to Resolve

Onsite Customer Engineer

Onsite FAPL Engineer

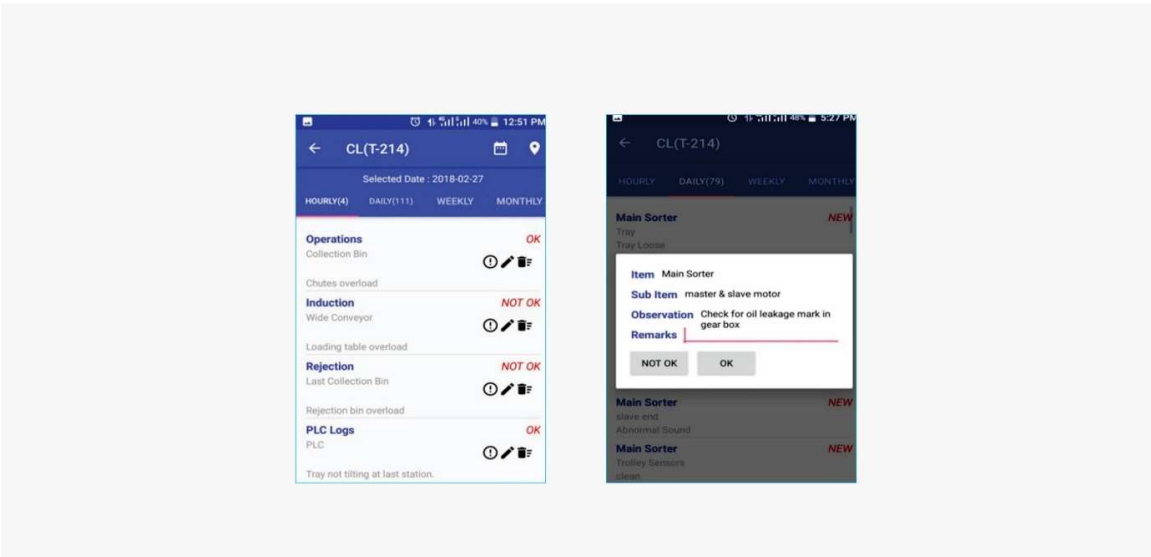
APPROVE ISSUE **OPEN ISSUES**

Search

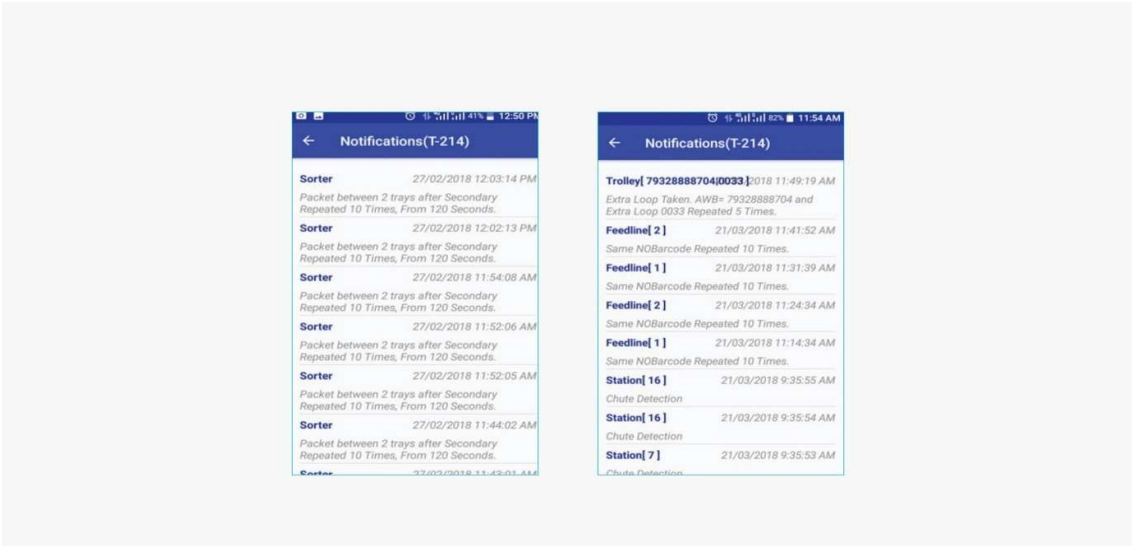
ID	Problem	T-214
1489	PTL Not Working	T-214
109	sorter stopped	T-214
107	Trolley Not Tilting	T-214
106	Trolley Not Tilting	T-214
96	Trolley Not Tilting	T-214
95	wrong sorting from feed lane-1	T-214

Problem PTL Not Working
Reason due to green light not blinking
Corrective Action programe done
Date 29/12/2017 12:00:00 AM
CANCEL **OK**

b. Preventive Maintenance Check List



c. Predictive Maintenance



d. Training Portal

Customer Management Product Management Machine Management Enginere & Training Ticket & Mail Problem Management Reports

Home Dashboard

Employee Training Admin Add New

Employee Name Training Type Search Select Page Size

Show 10 Scheduled Completed

Employee Name	Training Type	Subject	Media	Training Date To	Actual Start Date	Actual End Date	Setting
Shankar, S. S. S.	Product Training	TB Tray Series TB Tray Load Bottle Sorter		Week Chockery A. *			2 10 10
Shankar, S. S. S.	Product Training	System PTL Connected Belt Conveyor Belt Conveyor for HTH		Week Chockery A. *			2 10 10
Shankar, S. S. S.	Product Training	REJECTION AND JAM CONVEYOR SORTER SYSTEM		Week Chockery A. *			2 10 10
Shankar, S. S. S.	Product Training	TB Tray Series SHEEP SORTER Bottle Sorter PTL POPUP SORTER PTL		Week Chockery A. *			2 10 10
Shankar, S. S. S.	Product Training	VOL Custom Conveyor Check Wagon Belt Tuck Check Wagon Custom Belt Conveyor		Week Chockery A. *			2 10 10
Shankar, S. S. S.	Product Training	TB Tray Series SHEEP SORTER Bottle Sorter		Week Chockery A. *			2 10 10

Customer Management Product Management Machine Management Enginere & Training Ticket & Mail Problem Management Reports

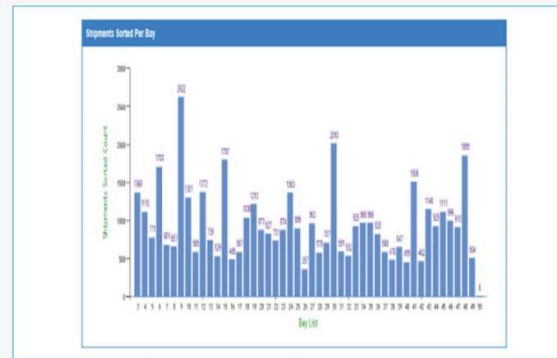
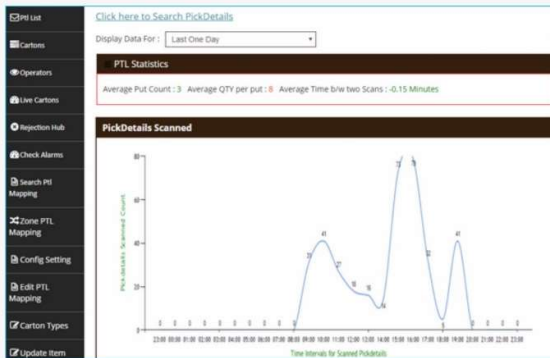
Home Dashboard

Training Material Master Admin Add New

Training Type Document Name Video Name Search Select Page Size

Training Type	Subject	Document Name	Training Document	Video Name	Training Video	Time Link	Setting
Component Training	KCH TRAINING	SOPAC MANUAL	download			2	10 10
Component Training	SFD TRAINING	POWERFEED AIR MANUAL	download			2	10 10
Component Training	PTL	PTL INSTRUCTION MANUAL	download			2	10 10
Component Training	PTL	OPERATIONAL MANUAL	download			2	10 10
Product Training	TB Tray Loop Sorter	T270 Secondary Sorting Manual	download			2	10 10
Maintenance & Troubleshooting	TB Tray Sorter	T274 MAINTENANCE MANUAL	download			2	10 10
Product Training	TB Tray Loop Sorter	T274 SECONDARY MANUAL	download			2	10 10
Product Training	TB Tray Loop Sorter	T274 REDUCTION LINE MANUAL	download			2	10 10
Department Procedure	Feed & Pallet Filling	Electrical-01	download			20 Min	10 10

e. Operational Analysis



19 Commercials

Price List- Summary

S.NO	Package	Price
1	Conveyors Package	28,34,72,928
2	Cross Belt Sorter Package	29,16,52,094
3	Destinations Package	3,83,87,721
4	Services Package	3,46,22,786
Total		64,80,90,529
"Business Cooperation Agreement" Discount for Delhivery		4%
Final Total		62,21,66,908

Break Up

1. Conveyors Package

S.NO	Product Group	Units	Qty	Total Price
1	PVC Belt Conveyor- 1200/1400 mm Width	mt	1872	9,74,73,581
2	PVC Conveyor Drive Units	Count	178	
3	Modular Belt Conveyor - 1200/1400 mm Width	mt	216.5	1,72,73,669
4	Modular Belt Conveyor Units	Count	15	
5	Telescopic Belt Conveyors Model- 7/21	Count	59	9,42,27,204
6	Singulators	Count	3	1,78,41,600
7	Swivel Wheel Divert Units	Count	3	33,55,278
8	30/60/90 Degree Curve Conveyor- Modular (1200/1400 mm Width)	Count	24	2,99,11,538
9	30/60/90 Degree Curve Conveyors- PVC (1200 mm Width)	Count	12	
10	Aligning Conveyors	Count	5	13,80,000
11	Idler Roller Conveyors (at Chutes)	mt	141	25,40,520
12	Angle Merge Conveyors	Count	3	8,23,627
13	Buffer Conveyors (prior to Sweddi)	Count	6	13,88,822
14	Control Package- Wiring + Cable Trays+ E-Stops and Pull Chords + PLC Control panels with HMI+ Tower Lamps and Hooters+ SCADA	Count		1,78,59,849
	Total			28,34,72,928

2. Cross belt Sorter package

S.NO	Product Group	Units	Qty	Total Price
1	Lower Deck CBS- Loop Length- 512.4 meters Number of Carriers- 436 Number of Belts- 872 2 X ECDS System 2 X Centring System 2 X Positioning System 1 X Loop Control System 1 X Power Distribution System 1 X Carrier Positioning System 1 X Data Transmission System Drive System Sorter Control System with HMI 1 X Additional VMS for Second Zone	Count	1	10,71,99,531
2	Upper Deck CBS- Loop Length- 512.4 meters Number of Carriers- 436 Number of Belts- 872 1 X ECDS System 1 X Centring System 1 X Positioning System 1 X Loop Control System 1 X Power Distribution System 1 X Carrier Positioning System 1 X Data Transmission System Drive System Sorter Control System with HMI	Count	1	10,50,90,835
3	Auto Inducts Comprising of- Buffer Conveyors Receiving Conveyors Spacing Conveyors MLG System Merge Conveyors Weighing Conveyor Control System with HMI	Count	6	6,19,15,906

4	Semi Auto Inducts Comprising of- Loading Conveyors Spacing Conveyors Weighing Conveyors MLG System Hand Held Barcode Scanner Weighing Conveyor Control System with HMI	Count	14	
5	Barcode Scanning and Dimensioning System- 5 Sided Scanning using 3 X SICK ICR 890+ Volume Scanning using 1 X SICK VMS 4200+ Associated Control System, Mirrors, Wiring and Aluminum Frame	Count	2	1,58,07,692
6	HAA Server Setup (please refer to the technical proposal for details)	Count	1	16,38,129
Total				29,16,52,094

3. Destination Package

S.NO	Chute Type	Description	Qty	Price
1	Sliding Chutes-	Inbound Highway Bag Segregation Chutes	32	3,30,67,721
2	Sliding Chutes	OB Chutes Feeding TBC	27	
3	Sliding Chutes	Auto Induct Reject Highway Chute	1	
4	Sliding Chutes	Lower Deck Auto feedline Reverse Chutes	2	
5	L Type Chutes	Upper Deck Auto Feedline Reverse Chute	4	
6	L Type Chutes	Main Sorter Collection Chutes	65	
7	L Type Chutes	Main Sorter Direct OB Chutes	28	
8	Rejection Chutes	Main Sorter Rejection Handling Chute	1	
9	Irregular Shipment Chutes	Upper Deck VDS- Exception Chute	1	
10	Irregular Shipment Chutes	Lower Deck VDS- Exception Chute	1	
Chutes I/O				
11	Chute Full Sensors		94	53,20,000.00
12	Tower lamps (2 colour)		94	
13	Push Button		94	
14	I/O for Chute full sensors		212	
Total				3,83,87,721

4. Services Package

S.NO	Service	Description	Qty	Price
1	Packaging and Forwarding	Wooden/Cardboard packing of system modules along with associated documentation and labelling for a safe transport to Client Site.	1	61,34,677
2	Project Management	A Dedicated Project Delivery group will be earmarked for the project comprising of - 1 X Project Engineer - 1 X Project Manager - 1 X Team leader This group will be supervised by Head of Projects and shall work very closely with your teams for a smooth execution of project and any risk mitigation. This also included preparation of Detailed Engineering Drawings, Layouts, Business requirement Documents, Manuals and Guides. This also includes travel to the site and your Head office on need basis This also includes coordinating with your various vendors, consultants, contractors etc	1	64,16,738
3	Installation and Commissioning	This includes the complete mechanical installation, electrical installation, Commissioning, Training, stabilization support and handover of the system to the client. During this phase, a team comprising of Site Manager + Electrical Engineers + Mechanical Engineers + PLC Engineers + Software Engineers + Store Manager + Safety Incharge will be stationed at your site and will supervise the satisfactory delivery of the project. We shall be hiring a reputed manpower contractor for executing this project. The Interstate Travel, Local travel, Stay, Food, daily allowances are all included in this	1	2,14,71,371
4	Software Services	This includes the Supply of Falcon's SortIT WCS Package customized to your requirement + Custom IT integration with your Host IT Systems	1	6,00,000
Total				3,46,22,786

The Total Price is to be understood

- Fright: Extra
- Including Packing and Forwarding
- Duty and Taxes Extra
- Installation and Commissioning: Mentioned above.

*Material Unloading & Laydown area is in customer scope.

*Scaffoldings, Man Lift etc. required for reaching the maintenance platforms shall be in customer scope

20 Payment Terms

Stage 1: 40% Advance along with Purchase Order

Stage 2: 50% along with 100% Taxes against Delivery of Material on Site on Pro-rata basis.

Stage 3: 10% after successful installation and commissioning.

21 Warranty Period

Falcon offered Systems come with a standard warranty of 12 months from the date of commissioning of the system or 15 months from the date of completion of dispatch of the materials whichever is earlier. End of commissioning is considered as the date on which the system is taken into operation for regular sorting activity. In case of delay in site readiness or lifting of materials from Falcon works, the 15-month period will be considered to start from the date of official communication of material readiness at Falcon end. The warranty include :

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

22 Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Scaffoldings, Man Lift etc. required for reaching the maintenance platforms.
- Air Compressor
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- TV Screens for Docks and Chutes
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.